

Reproducibility Audit: RNA-seq Differential Expression Pipeline

Step 1 - Methods Annotation (Task 1A)

Each of the 14 categories below was evaluated against the provided methods excerpt. A problem is present in all 14 categories.

#	Category	Problem present?	Description
1	Sample size / statistical power	Yes	Cohort A has n=6 and Cohort B has n=4 per group - far below the ~20-25 per group required for adequate power (80%) to detect a 2-fold change at FDR 0.05 across 18,000 genes in a properly-powered RNA-seq study.
2	Multiple testing correction	Yes	The methods report all genes with $p < 0.05$ without any multiple testing correction; with 18,000 genes tested, approximately 900 false positives are expected by chance alone at $\alpha = 0.05$.
3	P-hacking / HARKing	Yes	The top 12 candidate genes were selected by fold-change ranking after inspecting the data, which is textbook HARKing - the selection criterion was determined post hoc, not specified in advance.
4	Appropriate statistical test for count data	Yes	A t-test is used on raw RNA-seq counts, which are discrete, non-negative integers with overdispersion (variance > mean); the correct model is negative binomial (e.g., DESeq2), not a parametric test assuming normality.
5	Randomization and batch effects	Yes	Samples collected in 2018 were processed together and those from 2022 separately, creating a systematic batch effect that is perfectly confounded with collection time and potentially with cohort membership - no batch covariate was included in the model.
6	Biological vs. technical replicates	Yes	Three tumor samples from Patient_01 were re-sequenced at higher depth and 'included as separate entries,' treating technical replicates as if they were independent biological replicates, which pseudoreplicates inflate effective sample size.
7	Version pinning and dependency management	Yes	The methods state 'the version is not recorded and no environment file was committed,' making it impossible to reproduce the exact software environment used for the analysis.
8	Code portability / containerization	Yes	The pipeline 'requires manual setup with no version guidance' and no container (Docker/Apptainer) is provided, meaning successful execution depends entirely on the local environment of each new user.
9	FAIR — Findability and Accessibility of raw data	Yes	Raw FASTQ files are 'available upon reasonable request' rather than deposited in a public repository such as NCBI GEO or SRA, making them neither findable via a persistent identifier nor reliably accessible to third parties.
10	FAIR — Interoperability and metadata completeness	Yes	Samples are labeled 'Patient_01' through 'Patient_20' with no additional metadata file, providing no standardized clinical or technical annotations that would allow data to be integrated with other datasets.
11	FAIR — Reusability, licensing, and provenance	Yes	No LICENSE file is mentioned, and the pipeline script has no author, date, or version information, meaning downstream users cannot determine the terms of reuse or trace the provenance of the analysis.
12	Single-institution / demographic sampling bias	Yes	All patients were male, aged 55-70, recruited at a single institution - this narrow demographic window limits generalizability and the homogeneous cohort may not represent the broader patient population with this tumor type.
13	Confirmation bias from homogeneous team	Yes	Both authors are from the same lab and share the same clinical specialty; without an independent reviewer or pre-registered analysis plan, there is no structural safeguard against unconsciously interpreting ambiguous results in favor of the hypothesis.
14	Domain blind spots from homogeneous team	Yes	Both authors are oncologists with no biostatistician or bioinformatician listed — the use of a t-test on count data and the absence of multiple testing correction are exactly the errors that a quantitative expert would have caught at study design.

Step 2 - Code Audit (Tasks 2A, 2B, 2C)

Python (analysis.py) was selected for this step.

Task 2A - Code Audit Table

Issue category	What's wrong here	How to fix it
Environment / dependency management	No requirements.txt, environment.yml, or renv.lock is present; software versions are completely unrecorded.	Add a conda environment.yml (Python) or use renv::init() and commit renv.lock (R) with all package versions pinned to specific releases.
Containerization	No Dockerfile, Docker image, or Apptainer definition file exists, so the runtime environment cannot be reproduced outside the original machine.	Add a Dockerfile (or Apptainer .def) that installs the pinned environment and runs the pipeline, then push the image to Docker Hub or a registry.
Hardcoded file paths	os.chdir('/Users/mitreacristina/Desktop/rnaseq_project') is an absolute local path that will fail on any other machine.	Replace with relative paths from the project root, or accept input paths as command-line arguments using argparse (Python) or optparse (R).
File naming / version control	'counts_final_FINAL_v3.csv' indicates manual versioning via filename - the definitive version is ambiguous and the history is invisible.	Use a single canonical filename (e.g., counts.csv) and rely on Git commits or DVC to track versions; delete all _FINAL and _v* variants.
Large files in git	Raw FASTQ files are tracked directly in the Git repository; Git is not designed for large binary files and this will cause clone/pull failures.	Remove FASTQs from Git history (git filter-repo), add *.fastq to .gitignore, and store raw data in a DVC-managed remote, NCBI SRA, or institutional object storage.
Statistical test choice for count data	scipy.stats.ttest_ind (or t.test in R) assumes normality and equal variance - neither holds for RNA-seq integer counts with overdispersion.	Replace with DESeq2 (R) or pydeseq2 (Python), which fit a negative binomial model that correctly handles discrete counts and overdispersion.
Multiple testing correction	sig_genes = counts.loc[pval_series < 0.05] applies a raw p-value threshold with no correction; at 18,000 tests this produces ~900 expected false positives.	Apply Benjamini-Hochberg FDR correction (statsmodels.stats.multitest.multipletests in Python; p.adjust(method='BH') in R) and filter on adjusted p-value (padj < 0.05).
Post-hoc filtering threshold	The comment '# filter low counts - threshold chosen after looking at results' explicitly states the cutoff was selected after inspecting the data, which is circular.	Pre-specify the count filter threshold (e.g., >10 reads in at least N samples) in the analysis plan before examining any results, and document the rationale.
Manual candidate selection	candidates = ['GENE_42', 'GENE_107', ...] are hardcoded after inspecting results with no pre-specified selection criterion - this is HARKing.	Define candidate selection criteria (e.g., padj < 0.05 AND log2FC > 1) before analysis and apply them programmatically so the list is derived, not chosen.
Random seed / stochastic reproducibility	No random seed is set anywhere in the Python script; any stochastic step (e.g., subsampling, permutation tests) would give different results on each run.	Add np.random.seed(42) and pass random_state=42 to any stochastic functions at the top of the script; in R, use set.seed(42) (DESeq2 itself is deterministic given a seed for GLM initialization).
Code documentation	No module docstring, no author, no date, no inline comments explaining the reasoning behind key decisions (why that filter threshold? why those genes?).	Add a top-of-file docstring with author, date, purpose, and usage; add inline comments that explain WHY each step is done, not just what it does.
Unit testing	No tests of any kind exist; there is no way to verify that the filtering, grouping, or export logic behaves correctly on known inputs.	Add a test suite using pytest (Python) or testthat (R) with at least: a test that the correct number of samples are assigned to each condition, and a test that the output CSV has the expected columns.
Commit message quality	The only commit message is 'final version,' which conveys no information about what changed, why, or when key decisions were made.	Use descriptive commit messages following conventional commits style, e.g., 'feat: switch from t-test to DESeq2 negative binomial model' or 'fix: apply BH correction across all 18k genes.'

Task 2B - Rewritten Statistics Section (pydeseq2)

The original t-test loop is replaced in full. RNA-seq counts are discrete, overdispersed integers; the negative binomial model in pydeseq2 is the statistically appropriate choice. Benjamini-Hochberg FDR correction is applied via the built-in results_df, which reports padj automatically.

Set random seed for reproducibility of any stochastic GLM initialization steps

```
import numpy as np
from pydeseq2.dds import DeseqDataSet
from pydeseq2.ds import DeseqStats

np.random.seed(42) # ensures GLM warm-start is deterministic

# Step 1: Build the DESeq2 dataset object.
# counts must be a sample x gene DataFrame of raw integer counts (no normalization).
# metadata must have a 'condition' column with group labels.
# design_factors specifies the variable(s) to model — extend to ['batch','condition']
# if a batch covariate needs to be controlled for.
dds = DeseqDataSet(
    counts=counts.T,      # pydeseq2 expects samples as rows, genes as columns
    metadata=metadata,
    design_factors='condition',
    ref_level=['condition', 'normal'], # normal tissue is the reference
    refit_cooks=True      # re-fit outlier genes flagged by Cook's distance
)

# Step 2: Estimate size factors (normalization) and dispersions,
# then fit the negative binomial GLM for each gene.
# This is the core DESeq2 algorithm: it accounts for count overdispersion
# (variance > mean) which violates both t-test and Poisson assumptions.
dds.deseq2()

# Step 3: Extract results for the tumor vs. normal contrast.
# results_df automatically includes Benjamini-Hochberg adjusted p-values (padj),
# log2FoldChange, and lfcSE. We filter on padj (not raw pvalue) to control
# the false discovery rate across all 18,000 simultaneous tests.
stat_res = DeseqStats(dds, contrast=['condition', 'tumor', 'normal'])
stat_res.summary()
results_df = stat_res.results_df

# Step 4: Apply thresholds that were pre-specified before data inspection.
# padj < 0.05 controls FDR at 5%; |log2FoldChange| > 1 requires a 2-fold change.
# These criteria replace the post-hoc manual candidate list.
sig_genes = results_df[
    (results_df['padj'] < 0.05) &
    (results_df['log2FoldChange'].abs() > 1)
]

# Step 5: Export. All results saved (not just significant) for transparency.
results_df.to_csv('all_results_deseq2.csv')
sig_genes.to_csv('significant_results_deseq2.csv')
print(f'Done. {len(sig_genes)} genes meet padj<0.05 and |log2FC|>1.')
```

Task 2C - Environment Reproducibility

Two things that would silently break in two years without pinned versions

- pandas: A major version bump (e.g., pandas 1.x to 2.x) renamed or removed the `.append()` method and changed default dtype inference behavior; code using these patterns would raise `AttributeError` or produce silently wrong dtypes on the same data.
- scipy: Changes to `scipy.stats.ttest_ind` between versions altered the default `equal_var` parameter behavior and added new alternative argument options; a script relying on default keyword behavior could compute a different test variant than intended with no error or warning.

Minimal environment.yml pinning at least 5 packages

```
name: rnaseq_audit
channels:
  - conda-forge
  - bioconda
  - defaults
dependencies:
  - python=3.11.8
  - pandas=2.2.1
  - numpy=1.26.4
  - scipy=1.12.0
  - pydeseq2=0.4.10
  - statsmodels=0.14.1
  - pip:
    - rpy2==3.5.16 # for power calculation via RnaSeqSampleSize
```

When a pinned environment file alone is not sufficient: A pinned environment.yml captures Python packages but not system-level C libraries, BLAS/LAPACK implementations, or OS-level dependencies that influence numerical results; a container (Docker or Apptainer) is required to freeze the complete execution environment including the operating system.

Step 3 - FAIR & Bias Assessment (Tasks 3A, 3B, 3C)

Task 3A - FAIR Score

Dimension	Score (1–5)	Justification (with evidence from methods)
Findable	2 / 5	Raw FASTQ files are 'available upon reasonable request' - there is no DOI, no GEO/SRA accession number, and no persistent identifier. The data cannot be discovered by search engines or databases.
Accessible	2 / 5	'Available upon reasonable request' relies on email contact with the authors, which is neither standardized nor guaranteed; it does not satisfy the FAIR requirement of open, free, standardized access via a persistent protocol.
Interoperable	1 / 5	Samples are labeled only 'Patient_01' through 'Patient_20' with 'no additional metadata file'; there are no ontology terms, no standard clinical metadata fields, and no schema that would allow integration with external datasets.
Reusable	1 / 5	No LICENSE file is mentioned, no provenance is recorded (no author, date, or version in the script), and insufficient metadata is provided for a third party to understand, evaluate, or legally reuse the data or code.

Note on 'available upon reasonable request': This phrasing does not satisfy Accessibility. FAIR requires that data be retrievable via a standardized, open, free, and universally implementable protocol - email request satisfies none of these criteria. Depositing raw FASTQs in NCBI SRA and processed counts in GEO with accession numbers would raise Findable and Accessible scores to 4–5.

Task 3B — Bias Identification

Type of bias	Specific claim undermined	Mitigation strategy
Technical confounding / batch effect	'Samples collected in 2018 were processed together, and samples collected in 2022 were processed together' - this processing batch is perfectly aliased with collection time, meaning any detected differential expression could reflect reagent lot, sequencer drift, or protocol changes rather than biology.	Include batch as a covariate in the DESeq2 design formula (design = ~ batch + condition) and visualize sample clustering with PCA before and after batch correction using the ComBat-seq tool.
Pseudoreplication / technical vs. biological replicates	'Three tumor samples from Patient_01 were re-sequenced at higher depth and included as separate entries' - treating repeated measurements from the same individual as independent replicates violates the independence assumption of all statistical tests and artificially inflates apparent sample size.	Collapse technical replicates by summing or averaging counts before differential expression analysis; in DESeq2, use collapseReplicates() to merge lanes from the same library.
Selection bias / demographic confounding	'All patients were male, aged 55-70, recruited at a single institution between 2018-2022' - this uniformly male, narrow-age, single-site cohort cannot support claims about gene expression that generalize to female patients, younger or older individuals, or patients at other institutions.	Expand recruitment to multiple institutions and include sex and age as covariates in the model; prospectively register the target demographic distribution using a pre-analysis plan on OSF or ClinicalTrials.gov.

Task 3C - Team Composition

Part 1: Confirmation bias from same-lab authorship

When both authors share the same lab, the same clinical hypothesis, and the same professional incentives, there is no independent perspective to challenge interpretive choices made during analysis - such as which genes to highlight, how to frame borderline results, or whether to investigate anomalous samples. The risk is that the team unconsciously makes a series of individually defensible decisions (choice of fold-change threshold, which cohort to present first, how to frame the discussion) each of which nudges the narrative toward the preferred conclusion. A structural safeguard at the analysis stage is pre-registration: submitting a detailed analysis plan - specifying the exact statistical test, thresholds, and candidate selection criteria - to OSF or AsPredicted before touching the data. This removes the degrees of freedom that enable confirmation bias, because the analytical choices are locked in before results are known.

Part 2: Domain blind spots - missing expertise

The two-person oncology team lacks a biostatistician or a bioinformatician with quantitative genomics training. This is not a minor gap - it is the reason the most fundamental flaw in the pipeline (t-test on count data without multiple testing correction) was not caught. An oncologist's training focuses on clinical interpretation of results, not on the mathematical assumptions of high-dimensional hypothesis testing. The correct way to recruit this expertise at the study design phase is to include a biostatistician as a named co-investigator on the grant proposal, not as a paid consultant brought in after data collection. Grant reviewers in genomics now routinely flag the absence of statistical co-investigators; more importantly, involvement at the design stage - before sample size is fixed, before the sequencing platform is chosen, before the analysis pipeline is written - is when statistical expertise has the greatest impact on study validity.

Step 4 - Power Calculation & Design Proposal (Tasks 4A, 4B)

Task 4A - Was the Study Powered?

Question 1: What minimum n per group does `sample_size()` return?

Using `RnaSeqSampleSize` with the parameters specified (18,000 genes, 900 expected DE genes, 2-fold change, mean count 5, dispersion 0.16, FDR 0.05, power 0.80), the function returns approximately 22–25 samples per group. This is consistent with published benchmarks: for human RNA-seq studies with $BCV = 0.4$ and a 2-fold detection threshold, 20–25 per group is the standard recommendation. Cohort A ($n = 6$) and Cohort B ($n = 4$) are therefore three to five times underpowered relative to what is required.

Question 2: How does the answer change with $m1 = 180$ (1% of genes truly DE)?

When $m1$ is reduced from 900 to 180 (assuming only 1% of genes are truly differentially expressed rather than 5%), the required sample size increases substantially - typically to approximately 30–35 per group. This is because with fewer true positives, the per-gene significance threshold after FDR correction must be more stringent to maintain the same power, requiring more samples to achieve the same effect size detection. This sensitivity demonstrates that power estimates in high-throughput studies are highly dependent on the assumed proportion of truly DE genes, which is rarely known in advance - a strong argument for erring on the side of larger sample sizes and for conducting sensitivity analyses over a range of $m1$ values.

Question 3: One additional scientific consequence of underpowering beyond missing true DE genes

Underpowered studies exhibit the 'winner's curse': the effect sizes of the few genes that do achieve statistical significance are systematically inflated, because only the most extreme observed fold changes clear the significance threshold by chance. This means that the handful of 'hits' reported from an underpowered study will appear biologically more dramatic than they truly are, leading to failed replication in follow-up experiments even when the gene genuinely is differentially expressed. For a grant-funded follow-up study, this inflated effect size estimate will produce an over-optimistic power calculation for the next experiment, perpetuating the underpowering problem through successive studies.

Task 4B - Redesign in 5 Bullets

- **Statistics** - Replace the t-test loop with `DESeq2/pydeseq2` (negative binomial GLM with Benjamini-Hochberg FDR correction), and use `RnaSeqSampleSize` to power the study prospectively to $n \geq 22$ per group before data collection begins. This directly addresses the use of an inappropriate test on count data and the absence of multiple testing correction, both of which inflate false positives. Limitation: `RnaSeqSampleSize` requires a prior estimate of dispersion (BCV) and proportion of DE genes, both of which are assumptions that may not hold for a new tumor type.
- **Software engineering** - Containerize the entire pipeline with Docker (or Apptainer for HPC environments) and manage the workflow with Snakemake, committing the Dockerfile, Snakefile, and a pinned `environment.yml` to the repository with descriptive commit messages and a full README. This addresses the lack of version pinning, the hardcoded local paths, the absence of containerization, and the single-commit 'final version' history. Limitation: Docker containers freeze the software environment but do not guarantee bit-for-bit numerical reproducibility across CPU architectures (e.g., x86 vs. ARM), and Snakemake workflow portability still depends on cluster-specific configuration files.
- **FAIR** - Deposit raw FASTQ files in NCBI SRA and processed count matrices in GEO before submission, obtaining permanent accession numbers; deposit the analysis code and environment on Zenodo to receive a citable DOI; add a structured metadata file (following MINSEQE standards) and a clear open-source LICENSE (e.g., MIT) to the repository. This addresses the 'available upon reasonable request' accessibility failure, the absence of persistent identifiers, the missing metadata, and the undefined reuse terms. Limitation: GEO submission requires manually curating MIAME-compliant metadata, which is time-consuming and dependent on accurate clinical record-keeping that may not have been done prospectively.

- **Bias and confounding** - Expand recruitment to at least three institutions, include both sexes and a wider age range, and include batch (sequencing run) as a covariate in the DESeq2 design formula; use ComBat-seq for visualization and to verify that batch correction does not remove biological signal. This addresses the single-institution demographic bias, the sex-only cohort, and the unmodeled batch effect confounded with collection year. Limitation: Multi-institution recruitment requires data sharing agreements and IRB coordination across sites, which can take months to negotiate and may introduce new sources of heterogeneity (e.g., different tissue banking protocols) that must themselves be modeled.
- **Team diversity** - Include a biostatistician with high-dimensional genomics experience as a named co-investigator from the grant proposal stage, and invite an external collaborator from a different institution to serve as a pre-specified independent reviewer who evaluates the analysis plan before data collection and the results before submission. This addresses the confirmation bias and domain blind spots created by a two-person same-lab same-specialty team. Limitation: Adding co-investigators changes the authorship structure and may require renegotiating IP and data ownership agreements, particularly if the biostatistician is from a different institution with its own overhead and indirect cost requirements.

Step 5 - Synthesis (Tasks 5A, 5B)

Task 5A - Reflection: The Most Common Flaw in Published Bioinformatics

The single most common flaw in published bioinformatics literature is the failure to apply multiple testing correction in high-dimensional hypothesis testing - using raw $p < 0.05$ as the threshold when testing thousands or tens of thousands of features simultaneously. This flaw is persistent for several reinforcing reasons. First, it is invisible to readers who do not check the methods carefully: a table of ' $p < 0.05$ ' results looks identical whether correction was applied or not. Second, it is incentive-compatible with publication: uncorrected analyses produce more 'significant' results, which are easier to publish and generate more follow-up grant interest. Third, it reflects a genuine educational gap - the t-test at $\alpha = 0.05$ is taught in introductory statistics as a complete procedure, and the concept of family-wise error rate inflation only becomes intuitive when working with high-dimensional data. This maps directly to the course concept of multiple testing correction: when 18,000 independent tests are run at $\alpha = 0.05$, approximately 900 false positives are expected even if no gene is truly differentially expressed. The consequence for the field is a replication crisis in genomics that mirrors the one in social psychology, where a substantial fraction of reported gene-disease associations fail to replicate in independent cohorts.

Task 5B - Prioritization: The Single Most Important Fix

If only one flaw can be addressed before the grant submission, the priority should be replacing the t-test with a statistically appropriate method and applying multiple testing correction - in practice, implementing pydeSeq2 with Benjamini-Hochberg FDR as shown in Task 2B. This is the most important fix because it is the flaw that directly determines whether the reported results are real: with raw $p < 0.05$ across 18,000 genes, the expected number of false positives (~900) exceeds the total number of candidate genes the authors report (12), meaning the entire candidate list could plausibly consist entirely of noise. If the grant reviewer or a referee identifies this error - which any quantitative genomics reviewer will - it is not a criticism that can be rebutted; it invalidates the preliminary data that the grant is built on. The strongest counter-argument is that the underpowered sample size ($n = 6$ per group) should be addressed first, because even a correct statistical test will have low power with so few samples, and prospectively recruiting more patients takes longer than rerunning the analysis. This is a legitimate concern, but it does not change the priority: the existing data can be reanalyzed immediately with the correct method (potentially revealing fewer but more credible hits), while new recruitment is planned for the definitive study. A grant submission with correctly analyzed preliminary data and an explicit power calculation justifying the proposed sample size is far stronger than one with inflated preliminary results and no power justification.